

DO NOT WRITE ON TEST



**COMPUTER
PROGRAMMING
CONCEPTS
~OPEN EVENT~
(390)
STATE 2022**

TOTAL POINTS _____ **100 Points**

Test Time: 60 minutes

MULTIPLE CHOICE

Identify the choice that *best* completes the statement or answers the question.

1. ASCII Printable Characters such as letter, numbers and other symbols found on keyboards can be found in which range?
 - a. 00-31
 - b. 32-126
 - c. 128-191
 - d. 192-255
2. How many bytes does it take to store one ASCII character?
 - a. 1 Byte
 - b. 2 Bytes
 - c. 4 Bytes
 - d. 6 Bytes
3. The power, capabilities, and speed of a computer is determined by which of the following?
 - a. ROM
 - b. CPU
 - c. RAM
 - d. GPU
4. What is a data structure that stores and organizes information into tables?
 - a. Word Processor
 - b. Database
 - c. IDE
 - d. Spreadsheet
5. Which of the following is considered a low-level programming language?
 - a. Assembly Language
 - b. Java
 - c. Python
 - d. Scratch
6. Which of the following languages can a computer process without a previous transformation?
 - a. Machine Code
 - b. Python
 - c. FORTRAN
 - d. C++

7. Interpreters and Compilers are commonly known as what?
 - a. Translators
 - b. IDE's
 - c. Run Time Environments
 - d. SDK's
8. Which language is backwards compatible with C?
 - a. COBOL
 - b. C++
 - c. Java
 - d. BASIC
9. Which file type represents Java bytecode?
 - a. .java
 - b. .jar
 - c. .exe
 - d. .class
10. In logic programming, which of the following is one of the main concepts in creating the algorithm?
 - a. Logic
 - b. Machine Code
 - c. Abstraction
 - d. Flow Chart
11. Which of the following is a markup language?
 - a. XML
 - b. Scribe
 - c. HTML
 - d. All of these
12. Which of the following is a benefit of a Scripting Language?
 - a. Very compact compared to a compiled equivalent
 - b. Can run independent of a parent program
 - c. Gives full usage of a language
 - d. Helps make coding fast and simple
13. Which of the following best describes a minimal class declaration?
 - a. It cannot compile and it cannot execute
 - b. It can compile and execute
 - c. It can create new objects
 - d. It can compile but it cannot execute

14. Which of the following is a characteristic of a private data attributes?
 - a. There are restrictions to read/write only from members of the same class
 - b. There are restrictions to read only from members of the same class
 - c. There are restrictions to write only from members of the same class
 - d. There are restrictions to read only from members outside of the class
15. What is a characteristic of a “Get” method?
 - a. Read only methods that return a value
 - b. Read only void methods
 - c. Read/Write only methods that return a value
 - d. Read/Write only void methods
16. Which of the following best describes a class constructor?
 - a. There can be multiple constructors with the same identifier
 - b. A class can only have one constructor
 - c. A class must have two constructors, one of which is the default
 - d. Each constructor must be named differently
17. ArrayLists cannot do which of the following?
 - a. Store objects
 - b. Has its own methods
 - c. Store primitive data types
 - d. Store arrays
18. Which of the following is not a primitive data type in Java?
 - a. String
 - b. int
 - c. char
 - d. Boolean
19. What is the purpose of data encapsulation?
 - a. Place the data and methods in separate class so they do not corrupt each other
 - b. Place data and methods that access that data inside of the same class
 - c. Create all variables as private so no other classes can access them
 - d. Create all variables as public so all classes can access them
20. What is a characteristic of a “Set” method?
 - a. Read only methods
 - b. Read/Write methods
 - c. Read/Write only methods that return a value
 - d. None of these
21. Where must constructors be placed in a class declaration?
 - a. Their class declaration must be placed outside before they can be used
 - b. In the class declaration they need to be the first method
 - c. There is no required location for them to be placed in a class declaration
 - d. In the class declaration they need to be the last method

22. Inheritance can be best described by which of the following?
- a. Has-a relationship
 - b. Is-a relationship
 - c. Last In First Out relationship
 - d. Removal of data information
23. What is the name of a type of class that can use the features of its superclass?
- a. Inheritance Class
 - b. Composition Class
 - c. Utility Class
 - d. Subclass
24. Which method in Java will allow you to find out how many elements are contained in an ArrayList?
- a. size()
 - b. length
 - c. length ()
 - d. getSize()
25. Which of the following is a linear collection that can contain duplicated entries while also featuring the ability to access any element?
- a. Class
 - b. List
 - c. Set
 - d. 2D Array
26. Arrays can store which of the following?
- a. Objects
 - b. Arrays
 - c. Primitive data types
 - d. All of the above
27. Which of the ratings below would indicate the worst time for efficiency?
- a. $O(n)$
 - b. $O(\log n)$
 - c. $O(2^n)$
 - d. $O(n \log n)$
28. Based upon the condition of deleting the first element of an array data structure, which of the following would represent the time complexity?
- a. $O(1)$
 - b. $O(\log n)$
 - c. $O(n)$
 - d. $O(2^n)$

29. In which stage of the software development cycle would market research be performed?
- Designing
 - Testing
 - Planning
 - Identification
30. In which stage of the software development cycle would a requirements analysis be performed with stakeholders?
- Designing
 - Testing
 - Planning
 - Identification
31. Pick the definition below that best describes Structure Charts:
- Graphic representation of the statistics representing product failure
 - Graphic representation of the hierarchy of modules and the relationships between them
 - Drawings that represent the tolerance and dimensions of the computer infrastructure
 - Graphic that defines the preferred limits and fits for a software application
32. During which stage of the software development cycle do software engineers code the project?
- Designing
 - Testing
 - Planning
 - Implementation
33. What do software engineers use that provides comprehensive facilities during the development of software?
- IDE
 - API
 - SDK
 - JDK
34. What type of software development methodology is based upon an iterative process?
- Waterfall
 - RAD
 - Agile
 - Continuous Integration
35. Which of the following is not a programming paradigm?
- Procedural
 - Logical
 - Functional
 - Iterative

36. What shape in a flowchart is known as a terminal indicates the beginning and ending of a program or sub process?
- Pentagon
 - Diamond/ Rhombus
 - Oval/Ellipse
 - Circle
37. What shape in a flowchart indicates the input and output of data?
- Pentagon
 - Rhomboid/Parallelogram
 - Oval/Ellipse
 - Circle
38. Which of the following is a benefit of SQL?
- Allows the user to access a single record with one command
 - Allows the user to access multiple records with one command
 - Does not specify database behavior leaving implementations to decide how to behave
 - Semantic constructs of SQL is ambiguous
39. What type of database model is SQL?
- Entity-relationship model
 - Document model
 - Network model
 - Relational model
40. The rows in a table are known as what?
- Records
 - Fields
 - Queries
 - Scripts
41. Which of the following SQL statements selects the first, last and age columns from the “Employees” data table where the ages are greater than 50?
- SELECT first, last, age FROM Employees WHERE age > 49
 - SELECT first, last, age FROM Employees WHERE age >= 49
 - SELECT first, last, age FROM Employees WHERE age >= 51
 - SELECT first, last, age FROM Employees WHERE age >= 50
42. Which of the following SQL statements displays everyone’s first name and age columns from the “Employees” data table?
- SELECT first, age FROM Employees
 - SELECT * FROM Employees
 - SELECT f% a% FROM Employees
 - SELECT first, age FROM E%

43. What value is stored by the statement: `double d = (double)((int)91.0/(int)3.0)?`

- a. 30
- b. 30.0
- c. 33.333333333333332
- d. Error message

44. What value is stored by the statement: `double ans = 91.0/3?`

- a. 30
- b. 30.0
- c. 33.333333333333332
- d. Error message

45. What value is printed by the statement:

```
double a = 65.5;
char b = (char)((int)a);
double c = (int)b+5;
char d = (char) c;
print(d);
```

- a. C
- b. D
- c. E
- d. F

46. What value is printed by the statement:

```
int a = (int)(165.5 %100);
char b = (char)((int)a+2);
double c = (int)b+(a/3);
char d = (char) c;
print((char)((int)d-20));
```

- a. C
- b. D
- c. E
- d. F

47. What will be displayed by the method call *testRun (3,-10)*?

```
public static void testRun(int x, int y)
{
    if (x <= y)
    {
        print(x + " ");
        testRun (x+1, y);
    }
}
```

- a. 3 4 5 6 7 8 9 10
- b. 3 4 5 6 7 8 9
- c. 9 8 7 6 5 4 3
- d. Nothing will display

48. What will be displayed by the method call *testRun ('A', 'G')*?

```
public static void testRun(int x, int y)
{
    if (x <= y)
    {
        print(x + " ");
        testRun (x+1, y-1);
        print(y + " ");
    }
}
```

- a. 65 66 67 68 68 69 70 71 72 73 74 75
- b. 65 66 67 68 68 69 70 71 72 73 74
- c. 65 66 67 68 68 69 70 71 72 73
- d. 65 66 67 68 68 69 70 71 72

49. What will be displayed by the program segment below?

```
int twoD [ ][ ];
twoD = new int[3][4];

for(int a = 0; a < 3; a++)
{
    for(int b = 0; b < 4; b++)
        print(" " + (twoD[a][b]+b) + " ");
    print("\n");
}
println();
```

- a. 0 0 0 0 || 0 0 0 0 ||
- b. 0 1 2 3 || 0 1 2 3 ||
- c. 0 1 2 3 || 0 1 2 3 || 0 1 2 3 ||
- d. ArrayIndexOutOfBoundsException

50. When the program finishes running, what will be printed in the console?

```
char div = 'C';
int x = 100/div;
switch(div)
{
    case 'A': x += div;
    case 'B': x += 2;
    case 'C': x += 3;
    case 'D': x += 4;
}
print(x);
```

- a. 4
- b. 7
- c. 8
- d. 9